Open-Source Development

ITIC2262

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EXAMINE THE USE OF OPEN-SOURCE PRODUCT TO IMPROVE BUSINESS

Learning Outcomes

By the end of the session, students should be able to:

- Explain how OSS products are used in businesses.
- Identify business areas where OSS can be applied.
- Examine benefits and limitations of OSS adoption.
- Evaluate OSS products against business requirements.
- Recommend suitable OSS solutions for a business scenario.

What is an Open-Source Product?

- Software whose source code is available under an open-source licence.
- Users can generally use, study, modify and redistribute it according to the licence.
- Developed by individuals, communities, organizations, or companies.
- Examples: **Linux, Apache, PostgreSQL, MySQL, WordPress, Moodle, LibreOffice**.

Why Do Businesses Use Open-Source Products?

- Reduce software licensing costs
- Avoid vendor lock-in
- Customization
- Flexibility
- Community-driven innovation
- Integration with existing systems
- Access to a wide range of technologies

Business Areas Where OSS Can Be Used

- Operating Systems — Linux
- Web Hosting — Apache / Nginx
- Database — PostgreSQL / MySQL
- Website — WordPress
- E-Learning — Moodle
- Office Productivity — LibreOffice
- Collaboration — Nextcloud
- Business Management — Odoo
- Analytics — Metabase
- Development — Git / GitLab

OSS in Business Infrastructure

Example:

Users → Network → Linux Server → Web Server → Database → Business Application

OSS can support:

- Servers
- Networking
- Storage
- Databases
- Web applications
- Security

Improving Business Operations

Open-source products can improve:

- Operational efficiency
- Automation
- Communication
- Collaboration
- Data management
- Customer service
- Information sharing

Cost Reduction

Businesses may reduce:

- Software licence expenses
- Upgrade costs
- Infrastructure costs
- Vendor dependency

But consider:

- Implementation costs
- Training
- Technical support
- Maintenance
- Customization

Customization and Flexibility

OSS can allow organizations to:

- Modify source code where licensing permits.
- Add required functionality.
- Integrate with other systems.
- Adapt software to business processes.

Avoiding Vendor Lock-In

Proprietary approach:

Business → Vendor → Product → Vendor-specific ecosystem

OSS approach:

Business → Open technology → Multiple support/development options

Consider data portability, standards, migration and supplier independence.

Community Support

Open-source projects can receive support through:

- Documentation
- Forums
- GitHub/GitLab
- Tutorials
- Developer communities
- Bug reports
- Software updates

Community support quality varies between projects.

Professional / Commercial Support

Businesses may obtain:

- Consultancy
- Installation services
- Custom development
- Training
- Maintenance
- Technical support
- Managed services

Security Considerations

Potential advantages:

- Source-code transparency
- Rapid identification of vulnerabilities
- Security community
- Frequent updates for mature projects

Potential risks:

- Vulnerable dependencies
- Poorly maintained projects
- Incorrect configuration
- Delayed patching
- Unqualified customization

Scalability

OSS can support businesses from:

- Small businesses
- Medium enterprises
- Large organizations

Example progression:

Small business → Growing business → Enterprise

Infrastructure and OSS products should scale with business requirements.

Integration with Existing Systems

Businesses may need OSS to integrate with:

- Existing databases
- ERP systems
- Websites
- APIs
- Cloud platforms
- Authentication systems
- Network infrastructure

OSS and Business Innovation

OSS can help organizations:

- Experiment with new technologies.
- Build prototypes.
- Develop new services.
- Integrate emerging technologies.
- Reduce barriers to technology adoption.

Business Productivity Example

Small Business requirements:

- Office productivity
- File sharing
- Website
- Database
- Server

Possible OSS solutions:

Linux + LibreOffice + Nextcloud + WordPress + PostgreSQL

Measuring Business Improvement

OSS adoption can be evaluated using:

- Cost savings
- Productivity
- System availability
- Processing time
- Customer satisfaction
- Employee efficiency
- Scalability
- Maintenance requirements
- Return on investments

Advantages of OSS for Business

- Lower licensing costs
- Flexibility
- Customization
- Transparency
- Community innovation
- Interoperability
- Reduced vendor dependency
- Wide technology availability

Challenges of OSS Adoption

- Technical expertise required
- Training requirements
- Support availability
- Compatibility issues
- Maintenance
- Security management
- Licence compliance
- Project abandonment risk

OSS Licence Considerations

Open source does not mean “no licence.”

Examples:

- GPL
- LGPL
- MIT
- Apache License
- BSD licences

Businesses must understand modification, distribution, attribution, copyright and licence compatibility requirements.

Selecting an OSS Product

A business should evaluate:

Business Requirement → Features → Licence → Security → Community → Support → Cost → Scalability → Integration

OSS Evaluation Matrix

- Functionality — Does it meet business requirements?
- Cost — What are total implementation costs?
- Security — Is it actively maintained?
- Support — Is professional/community support available?
- Scalability — Can it grow with the business?
- Integration — Can it work with existing systems?
- Licence — Is the licence suitable?
- Usability — Can employees use it effectively?

Total Cost of Ownership

Do not consider only the purchase price.

TCO includes:

- Software
- Hardware
- Installation
- Configuration
- Training
- Support
- Maintenance
- Upgrades
- Security

OSS vs Proprietary Software

Open Source:

- source code generally available
- licence varies
- often high customization
- potentially lower vendor dependency
- community/commercial support

Proprietary:

- source code usually closed
- vendor-controlled licensing
- customization may be restricted
- vendor/partner support is common

Actual cost and suitability depend on the specific product and business requirements.

When OSS May Not Be Suitable

OSS may not be appropriate when:

- Required functionality is unavailable.
- Skilled technical staff are unavailable.
- Vendor-certified support is mandatory.
- Integration requirements cannot be met.
- Licence requirements conflict with the business model.
- Total cost of ownership is not favourable.

Business Decision Framework

Identify Business Problem

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Define Requirements

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Identify OSS Products

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Evaluate Alternatives

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Pilot / Test

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Security & Licence Review

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Calculate TCO

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Implement

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Monitor Business Benefits

THANK YOU

Case Study: E-Commerce Business

Scenario:

A small online business wants to develop an e-commerce platform.

Requirements:

- Website
- Product catalogue
- Database
- Customer management
- Secure hosting
- Reporting

Task:

Identify suitable OSS products and explain how they could improve the business.

Case Study Analysis

Students should examine:

1. Business problem
2. Required functionality
3. Selected OSS product
4. Features
5. Cost considerations
6. Customization
7. Security
8. Technical support
9. Scalability
10. Business benefits

Practical Activity — 15 Minutes

Group Activity

A company has 50 employees and wants to reduce software costs while improving collaboration.

Propose OSS products for:

1. Office productivity
2. File sharing
3. Website
4. Database
5. Operating system

Explain how each product could improve the business.

Discussion Questions

1. Does open source always mean free?
2. How can OSS reduce business costs?
3. How can OSS increase business flexibility?
4. What are the security risks?
5. Why is licence compliance important?
6. What factors should a business consider before adoption?
7. Can OSS completely eliminate proprietary software?

Assignment Connection

Assignment: Use of Open-Source Products in Different Applications

Students can apply today's concepts by:

- Selecting an OSS product.
- Identifying its application area.
- Explaining its major features.
- Examining its business benefits.
- Identifying limitations.
- Providing a suitable business use case.